Patrick Pataky

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Patrick-Pataky

Master student specializing in Computer Architecture and Digital Design. Experienced in developing custom RTL (Verilog/SystemVerilog) and embedded applications. Looking for an Embedded Systems and FPGA internship to contribute to hardware/software integration.

Education

- **Ecole polytechnique fédérale de Lausanne (EPFL)** *2024-Present*
  Master in Computer Science (with Cyber Security Specialization)
- **Ecole polytechnique fédérale de Lausanne (EPFL)** *2021 - 2024*
  Bachelor in Computer Science

Technical Skills and Interests

Languages & Architectures: Verilog, VHDL, Java, C/C++, Scala, Python, RTL, ARM, RISC-V, x86, CUDA, Bash
Tools & Frameworks: Vivado, FPGA (AMD Xilinx/Intel), Linux, Quartus, Yosys, QuestaSim, Git, GDB
Areas of Interest: Computer Architecture, Embedded Systems, Hardware Security, Compilers, HLS, Parallelism, Low-Level Programming, Accelerators, Hardware-Software Interactions, Algorithms, Data Structures, Compilers
Languages Spoken: English (fluent), French (native), Slovak (basic)

Personal Projects

- **X-HEEP RISC-V MCU: FPGA Verification and Flash Cache Design** *Spring 2026 (In Progress)*
  Semester Project at the Embedded Systems Laboratory (ESL), EPFL
  - **Hardware Verification:** Verifying a Flash Controller on a Xilinx FPGA for the open-source X-HEEP RISC-V MCU, utilizing Integrated Logic Analyzers (ILA) to debug hardware and software issues across I/O operations.
  - **RTL Design & Implementation:** Designing a Write-Back Cache in SystemVerilog (holding 1 to 4 sectors) to optimize memory access latency.
  - **Testbench Development:** Developing comprehensive C test applications to stress-test the hardware controller and identify functional bugs. Usage of TCL scripts to automate simulations in QuestaSim.
  - **Keywords:** SystemVerilog, RTL Design, FPGA Verification, C, Hardware Debugging (ILA), Testbenches, QuestaSim, TCL, Vivado, Hardware-Software Co-Design
- **FPGA Snake Game Hardware Accelerator** *Fall 2024*
  Course project from Embedded system design (CS-476)
  - **RTL Design & Implementation:** Developed a custom hardware accelerator from scratch on the Gecko 5 Education Board FPGA to offload graphical rendering from the CPU.
  - **Analytical Problem-Solving:** Achieved a 50x speedup over the software implementation, by profiling C code to identify critical paths and migrating them to the FPGA.
  - **Professor:** Prof. Dr. Theo Kluter
  - **Keywords:** Verilog, RTL implementation, Yosys, FPGA, Hardware Accelerator, Embedded Systems
- **Nintendo DS Multiplayer Game** *Fall 2023*
  Course project from EE-310: Microprogrammed Embedded Systems
  - **Embedded Systems & Real-Time Constraints:** Developed a multiplayer "Overcooked"-type game on the Nintendo DS, optimizing memory usage and tile-based rendering while handling input and animations on resource-constrained hardware.
  - **Professor:** Prof. Dr. David Atienza
  - **Keywords:** Embedded Systems, C, Tiling, RAM Optimization, DMA, Debugging, Analytical Skills

Experience

- **EPFL** *Fall 2023-Present*
  Teaching Assistant for courses with 300+ students
  - **Courses**: CS-173 (Fundamentals of Digital Systems), CS-200 (Computer Architecture), CS-302 (Parallelism and Concurrency in Software), CS-470 (Advanced Computer Architecture)
  - **Mentored 30–40 students weekly** providing guidance on digital logic, processor design, and programming labs.
  - **Debugged student implementations** in real-time for key projects, including designing and testing a simple in-order multicycle CPU from scratch in Verilog and writing a cycle-accurate C simulator of an out-of-order CPU.
- **Abalance SA, Neuchâtel (Switzerland)** *August 2023-September 2023*
  IT Developer / Infrastructure Developer
  - **Server installation**: Installed and configured Linux servers for a 10-person environment, and reduced manual setup time by 90% by automating repeatable steps with shell scripts.
  - **Maintenance**: Performed regular maintenance tasks to ensure network stability and software updates.